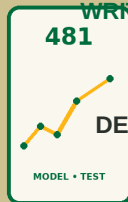
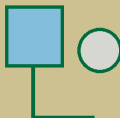
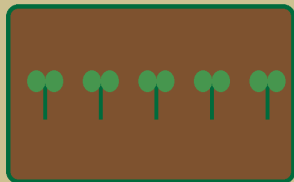


Write an Engineering Problem Statement



WRITE AN ENGINEERING PROBLEM STATEMENT

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

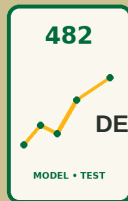
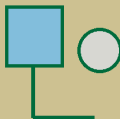
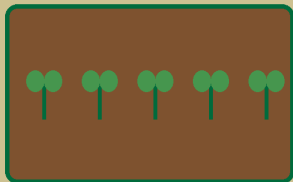
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Translate Stakeholder Needs



TRANSLATE STAKEHOLDER NEEDS

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

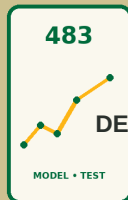
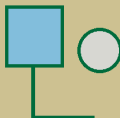
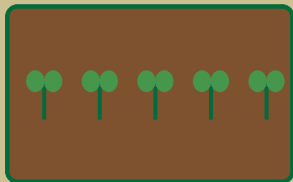
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Define Design Requirements



DEFINE DESIGN REQUIREMENTS

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

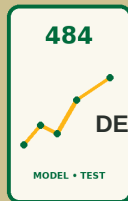
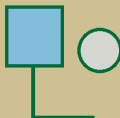
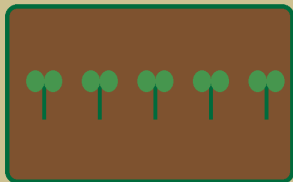
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Identify Design Variables



IDENTIFY DESIGN VARIABLES

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

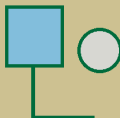
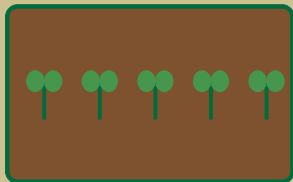
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Build a Decision Matrix



BUILD A DECISION MATRIX

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

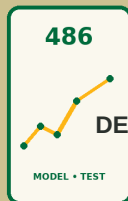
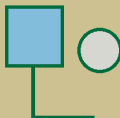
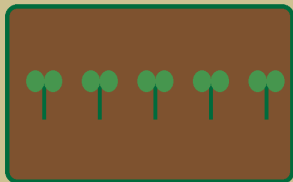
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Analyze Design Trade-Offs



ANALYZE DESIGN TRADE-OFFS

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

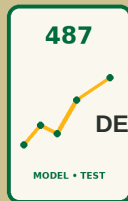
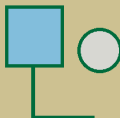
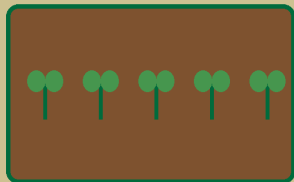
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Develop a Prototype Test Plan



DEVELOP A PROTOTYPE TEST PLAN

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

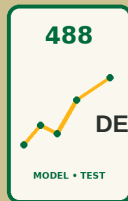
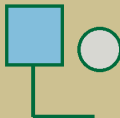
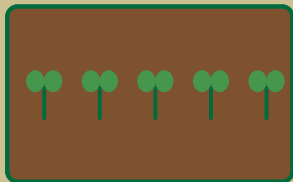
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Create a Technical Risk Register



CREATE A TECHNICAL RISK REGISTER

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

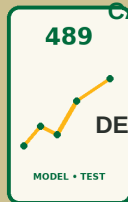
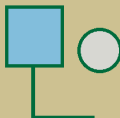
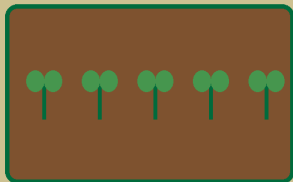
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Calculate a Techno-Economic Metric



CALCULATE A TECHNO-ECONOMIC METRIC

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

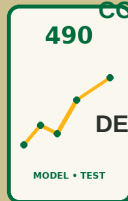
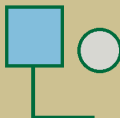
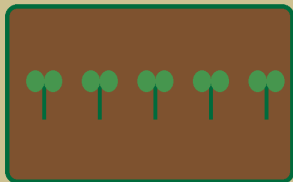
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Conduct an Engineering Design Review



CONDUCT AN ENGINEERING DESIGN REVIEW

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE